

SAMMIE KATT

PERSONAL INFORMATION

<i>Location</i>	Helsinki, Finland
<i>Citizenship</i>	Dutch
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EDUCATION

<i>Postdoc</i>	<i>2023-Present</i>	Northeastern University Postdoctoral researcher with Dr. Samuel Kaski with a focus on artificial theory of mind for human-in-the-loop settings.
<i>Computer Science PhD</i>		Research assistant of Dr. Amato with a focus on model-based Bayesian reinforcement learning in partially observable environments.
	<i>2015-2016</i>	The University of New Hampshire Research and teaching assistant (course on <i>Probabilistic AI and Machine learning</i>) for Dr. Amato. Research focused on Bayes-Adaptive Partially Observable Markov Decision Processes.
<i>Masters of Artificial Intelligence</i>	<i>2012-2017</i>	The University of Amsterdam Track: <i>Machine learning</i> Thesis: Efficient Bayesian Learning in Factored Partially Observable Environments. Adviser: Dr. F.A. Oliehoek.
<i>Bachelor Beta-Gamma</i>	<i>2009-2012</i>	The University of Amsterdam Major: <i>Artificial Intelligence</i> Interdisciplinary research Thesis: KBK-evac-program: the development of an artificially intelligent evacuation program. Artificial Intelligence Thesis: Introducing Movement and Animations to Virtual Victim in USARSim.

RESEARCH

Publications

<i>2025, AAAI</i>	Zhu, Y., Katt, S. , & Kaski, S. (2025). More Than Irrational: Modeling Belief-Biased Agents. In AAAI Conference on Artificial Intelligence. Association for the Advancement of Artificial Intelligence.
<i>2025, NeurIPS Workshop</i>	Katt, S., & Kaski, S. (2025). Artificial Theory of Mind in Human-in-the-Loop. In NeurIPS Workshop on Theory of Mind in Human-in-the-Loop.
<i>2025, TMLR</i>	Starre, R. A., Katt, S. , Çelikok, M. M., Loog, M., & Oliehoek, F. A. (2025). Abstraction for Bayesian Reinforcement Learning in Factored POMDPs. In Transactions on Machine Learning Research.
<i>2025, IROS</i>	Nguyen, H., Katt, S. , Xiao, Y., & Amato, C. (2023). On-robot Bayesian reinforcement learning for POMDPs. In 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) (pp. 9480-9487). IEEE.
<i>2022, AAMAS</i>	Katt, S. , Nguyen, H., Oliehoek, F. A., & Amato, C. (2022). BADDR: Bayes-Adaptive Deep Dropout RL for POMDPs. In <i>Proceedings of the 21th International Conference on Autonomous Agents and MultiAgent Systems</i> .

2019, AAMAS	Katt, S. , Oliehoek, F. A., & Amato, C. (2019). Bayesian Reinforcement Learning in Factored POMDPs. <i>In Proceedings of the 18th International Conference on Autonomous Agents and MultiAgent Systems.</i>
2019, ICRA	Xiao, Y., Katt, S. , ten Pas, A., Chen, S., & Amato, C (2019). Online Planning for Target Object Search in Clutter under Partial Observability. <i>International Conference on Robotics and Automation.</i>
2017, ICML	Katt, S. , Oliehoek, F. A., & Amato, C. (2017). Learning in POMDPs with Monte Carlo tree search. <i>In Proceedings of the 34th International Conference on Machine Learning-Volume 70.</i>
2016, SOCS	Cserna, B., Bogochow, M., Chambers, S., Tremblay, M., Katt, S. , & Ruml, W. (2016). Anytime versus real-time heuristic search for on-line planning. <i>In Ninth annual symposium on combinatorial search.</i>

Misc services

Organizing	Organizer of the 2024 AAAI Bridge Program on Collaborative AI and Modeling of Humans .
Reviewing	Several years of reviewing for most top-tier AI conference, including ICML (201-2025), NeurIPS (2023, 2024), AISTats (2025, 2026), and AAAI (2023-2026).

WORK EXPERIENCE

2020, Spring	Aptiv, Boston
Research Scientist Intern	Investigated self-driving car motion planning models conditioned on predictions of other vehicles. Designed and implemented a prediction-based planner in c++ capable of lane-following and lane-switching while considering the expected responses from other drivers. Developed, trained and tested a motion prediction network in python conditioned on other agent's predicted trajectories. This work generated a patent.

AWARDS

2013	Commendation for High Achievement
University of Queensland Exchange	Worked on a postgraduate project on modeling Total Artificial Hearts and took the classes 'Models of Software Systems', 'Systems Engineering' and 'Introduction to Web Design'. The grades and achievements during this exchange resulted in a Dean of Academic Programs' Commendation for High Achievement.
2012	Finalist, interdisciplinary research symposium
Institute of Interdisciplinary Studies, Bachelor	Reached the finals of the interdisciplinary research symposium. The group project resulted in an artificially intelligent 'KBK-evac-program', a building evacuation program. Contributed by designing the artificial intelligence aspect and the actual implementation of the software.

PROGRAMMING LANGUAGES & SOFTWARE

c++ · python · pytorch · tensorflow

LANGUAGES

DUTCH · Native

ENGLISH · Fluent